Technische Universität Braunschweig



Dislocations and Defects in Crystals

Lecture: Semiconductors

Source: Introduction to Dislocations (D. Hull and D. J. Bacon)

S. Haghi

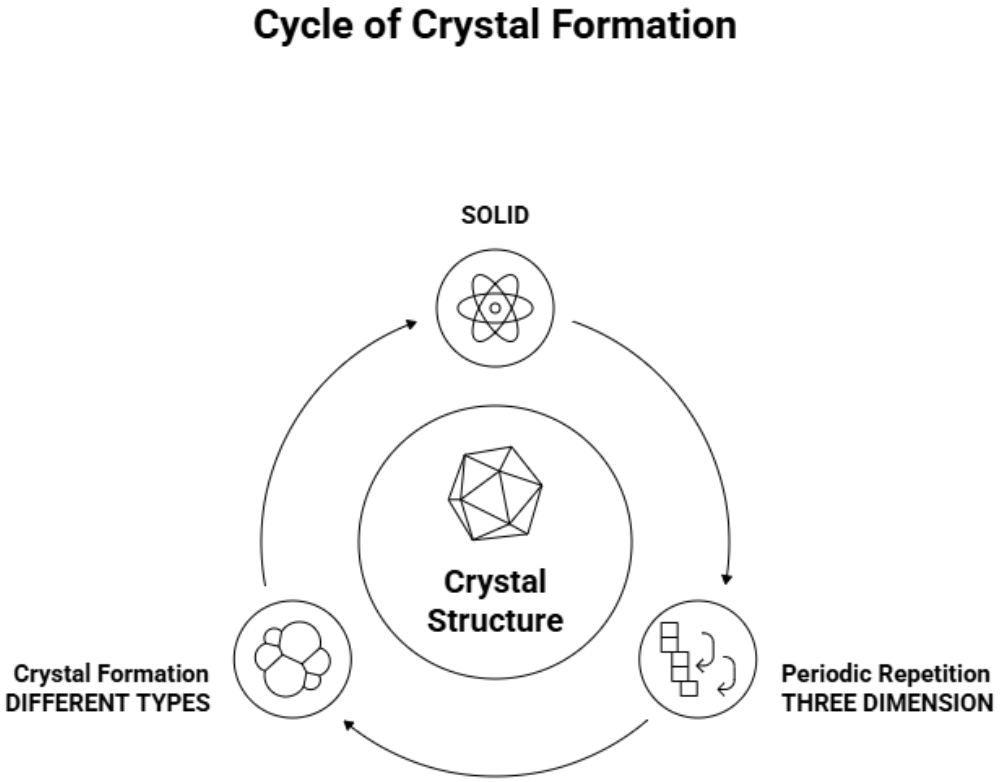
Prof. Andreas Waag

24.06.2025

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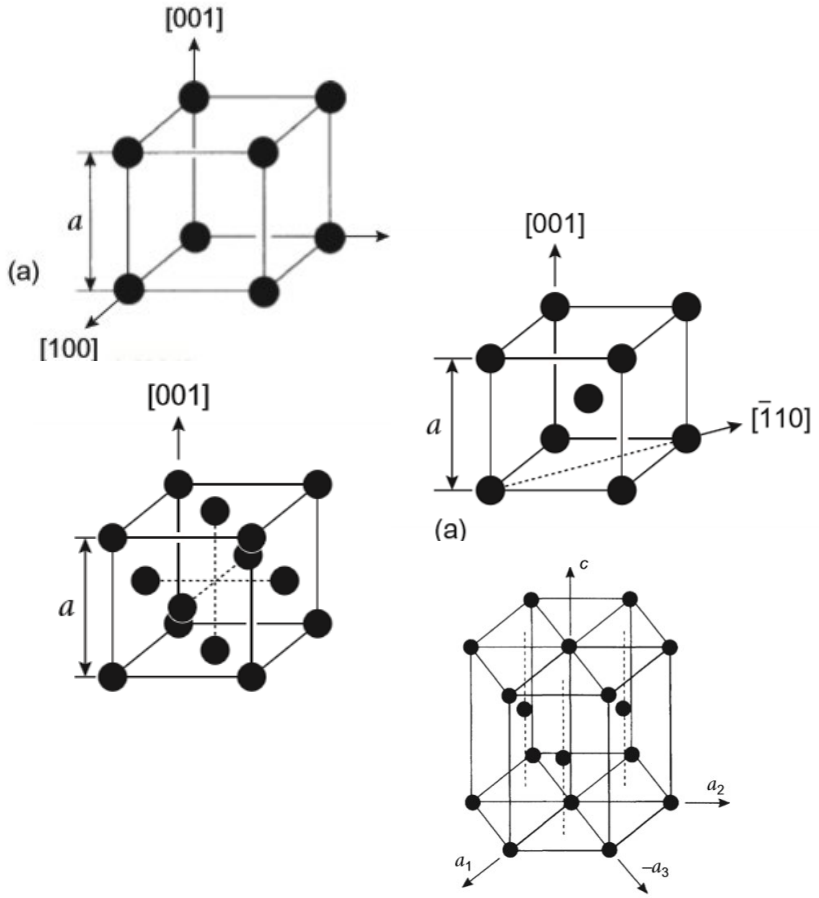
Crystals
What is the crystal?



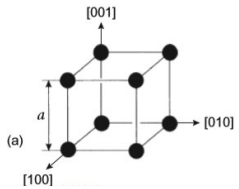
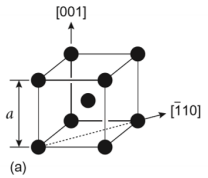
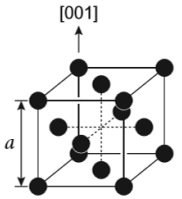
A **crystal** is defined as a solid in which its constituent atoms are **arranged in a pattern that repeats itself periodically in three dimensions**. This repeating arrangement describes the crystal structure.

What are the different types of crystals?

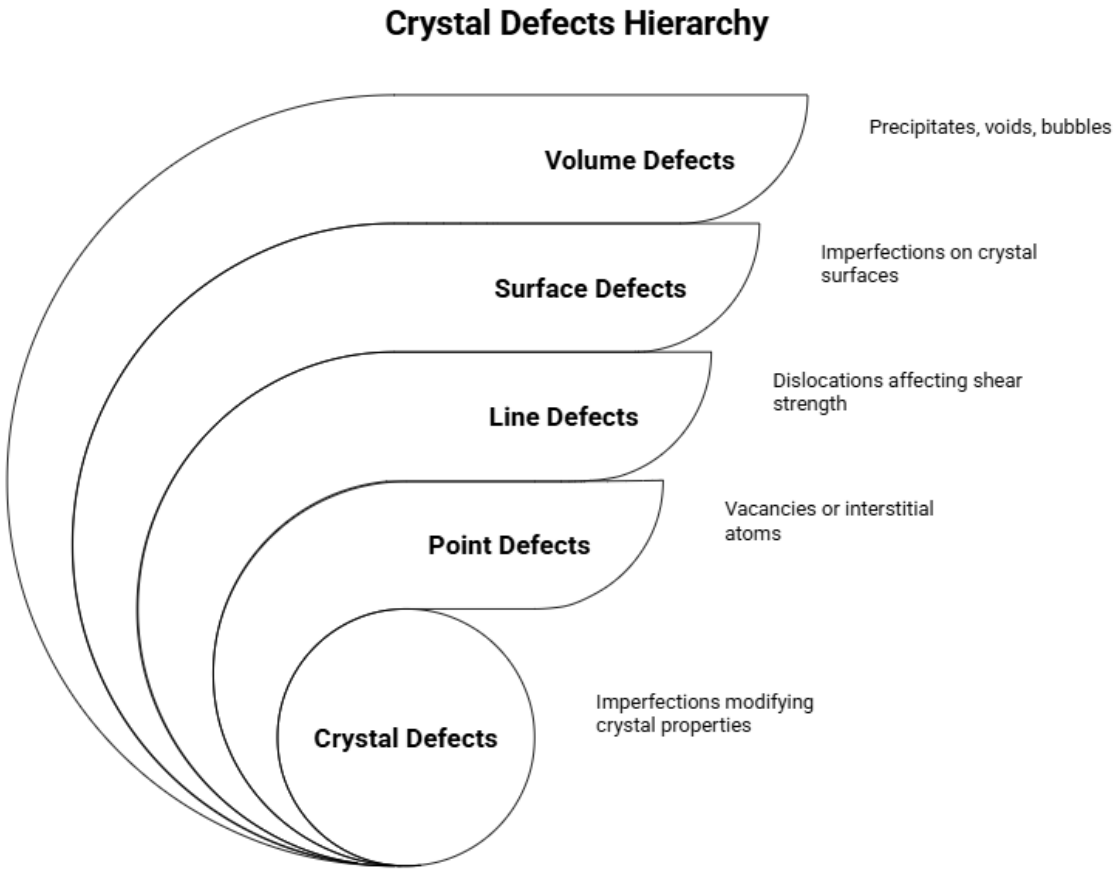
- **Simple Cubic (SC)**
- **Body Centered Cubic (BCC)**
- **Face Centered Cubic (FCC)**
- **Close-Packed Hexagonal (CPH)**



Crystal Structure Comparison

	 Simple Cubic (SC)	 Body-Centered Cubic (BCC)	 Face-Centered Cubic (FCC)
Atomic Arrangement	Corners only	Corners and center	Corners and face centers
Atom Touching	001> directions	111> directions	011> directions
Real-world Occurrence	No real crystals	Common for metals	Common for metals
Stacking Sequence	AAAA....	ABABAB...., ABCDEFAB....	ABABAB...., ABCABC....

DEFECTS
What is the Defect?

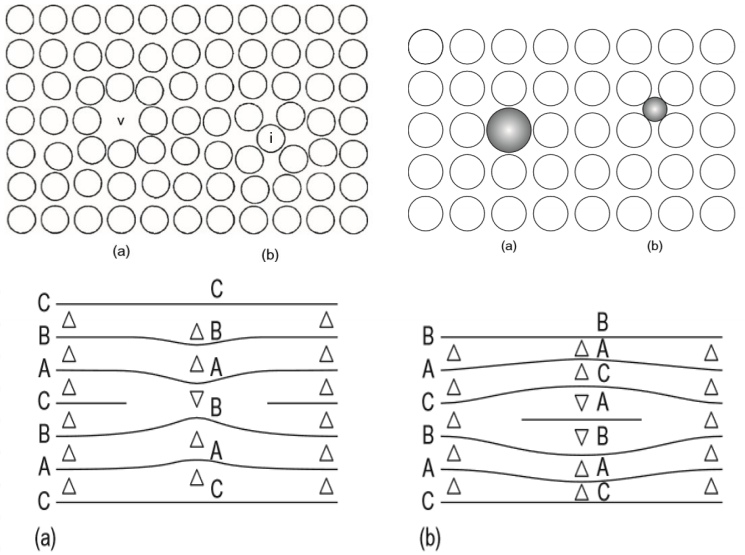


Point defects(localized disturbances)

Line defects(deformed)

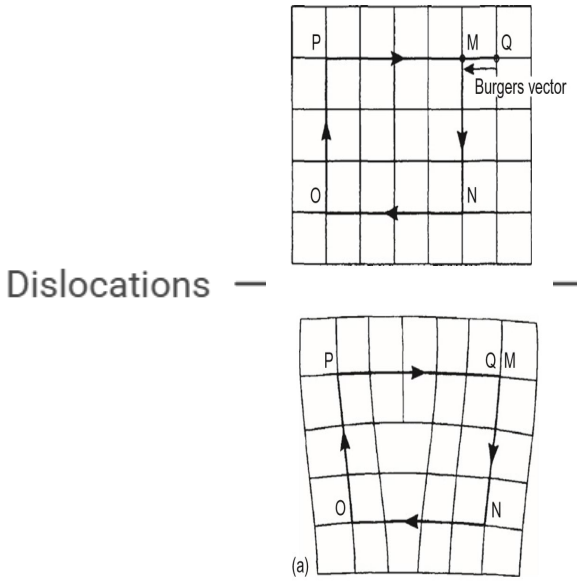
Surface defects(Two-dimensional imperfections)

Volume defects(Three-Dimensional imperfections)



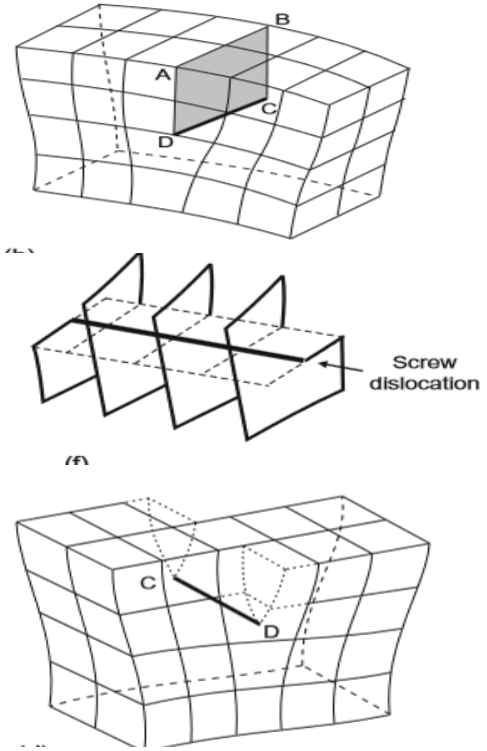
DISLOCATIONS
What are the Dislocations?

- Dislocations are an important class of **line defects** in crystalline solids.

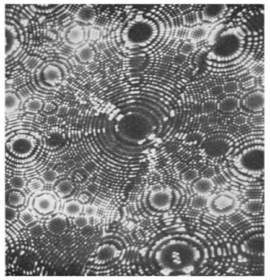


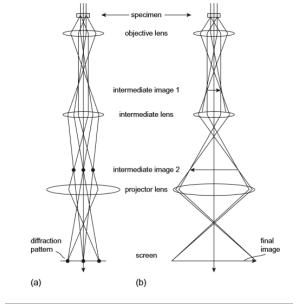
Dislocations → Burgers Vector → Edge Dislocation • **Normal**; Screw Dislocation • **Parallel**; Mixed Dislocation • **Both**

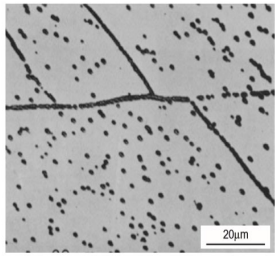
Burgers vector

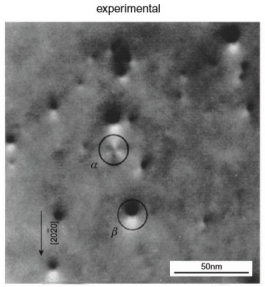


OBSERVATION





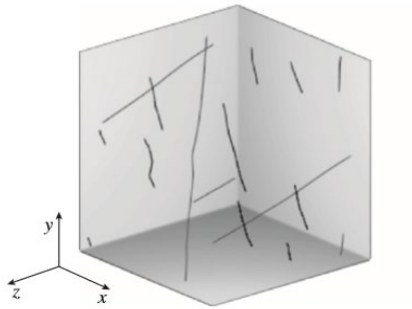




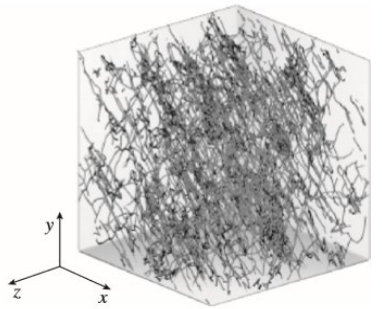
Characteristic	Atom Probe Tomography	Transmission Electron Microscopy	X-ray Diffraction Topography	Gas Ion Bombardment
Resolution	Atomic-level	High resolution	Lower resolution	Lowest resolution
Object	3D chemical distribution	Dislocation distribution	Crystal plane bending	Dislocation surface points
Principle	three dimensional field density	Electron transmission	X-RAY transmission	Surface etching

SIMULATION

Simulation plays
a **crucial and multifaceted role in understanding dislocations in semiconductors**,
often providing insights into phenomena that are **difficult or impossible** to observe directly at the atomic scale



(a)



(b)

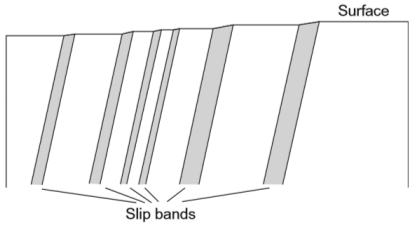
	Atomic-Scale	Dislocation Dynamics
Scale of Study	Up to 100 nm	Up to 100 μm
Underlying Principles	Atomic interactions	Elastic continuum
Type of Information	Atomic structure	Dislocation movement
Computational Demands	Very demanding	More efficient

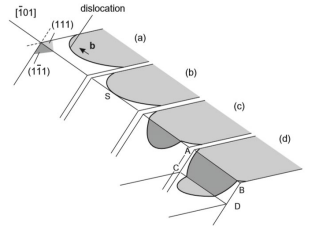
Movment

Dislocation movement is fundamental to **understanding the deformation** of crystalline solids and is significantly influenced by various factors, including:

temperature and
the surrounding environment.

There are two primary
Types of dislocation movement:
1. Glide
2. Climb





	Glide	Climb
Definition	Movement within slip plane	Movement out of slip plane
Mechanism	Displacement of atoms on plane	creation of vacancies
Slip Plane	Occurs on specific planes	different planes
Dislocation Type	available for Edge dislocations	available for screw dislocations
Driving Force	pressure on slip plane	vacancies suction
Temperature	not important	higher temprature

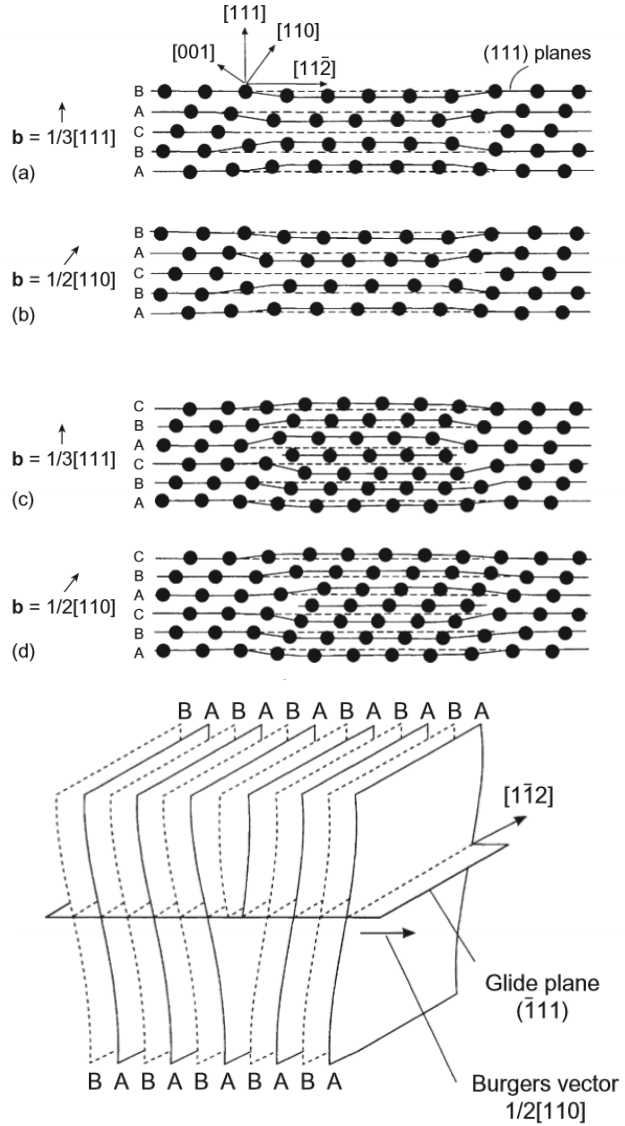
Dislocations in FCC

- A crystal structure where atoms are located at each corner and the center of all cube faces. Common in ductile metals like copper and aluminum, they exhibit easy slip on specific planes.

(e.g., Cu, Al, Ni)

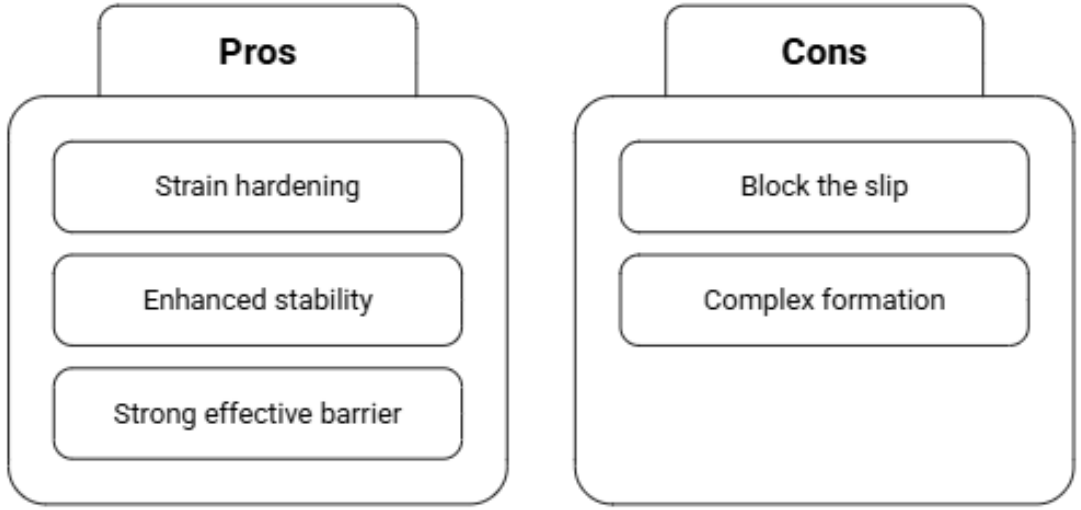
•Slip occurs on **{111} planes** in **<110> directions**
•Dislocations **dissociate into Shockley partials** → separated by a stacking fault
•**Cross-slip is possible**, but requires partials to recombine (thermal activation)

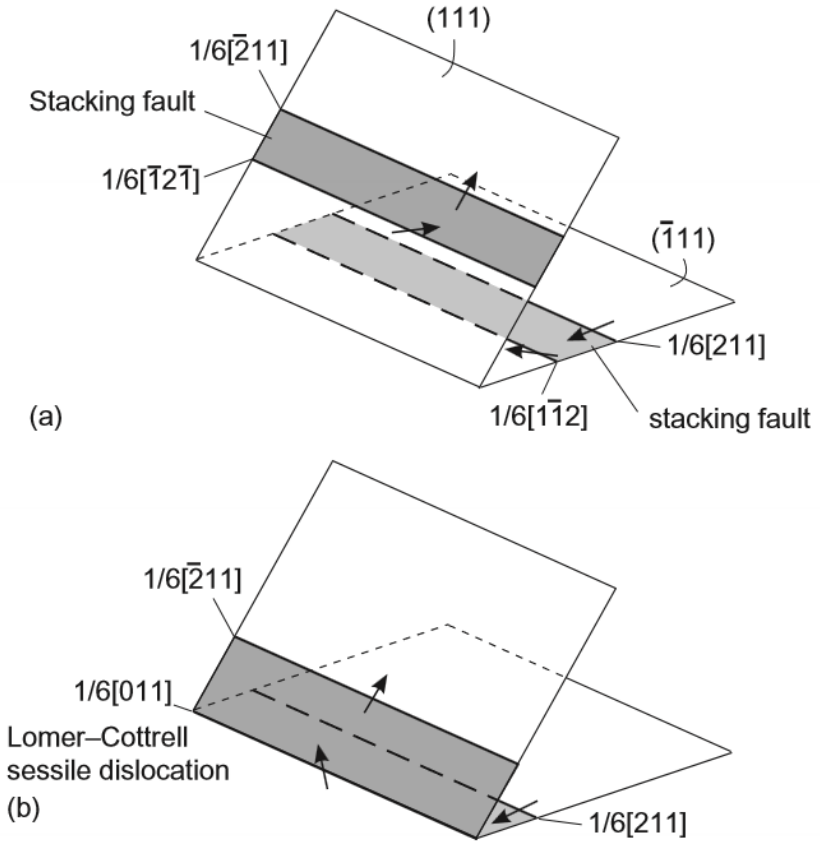
•**Work hardening** via:
- **Lomer-Cottrell locks** (sessile dislocations)



Lomer-Cottrell lock

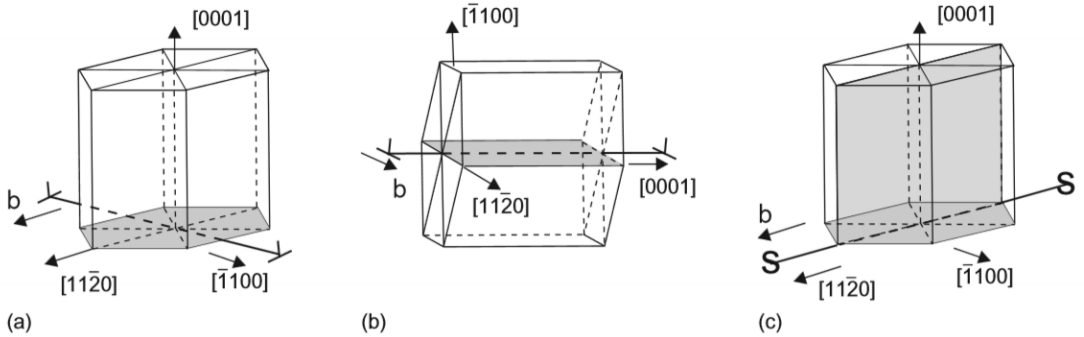
Defenition:
Strong
Connection of
Immobile and partial dislocations **in FCC**





Dislocations in HCP

- Crystals with a hexagonal unit cell, exemplified by metals such as magnesium and titanium. Their deformation behavior is influenced by the c/a ratio and preferred slip systems.



HCP Crystals (e.g., Mg, Ti, Zn)

•Slip depends on the material:

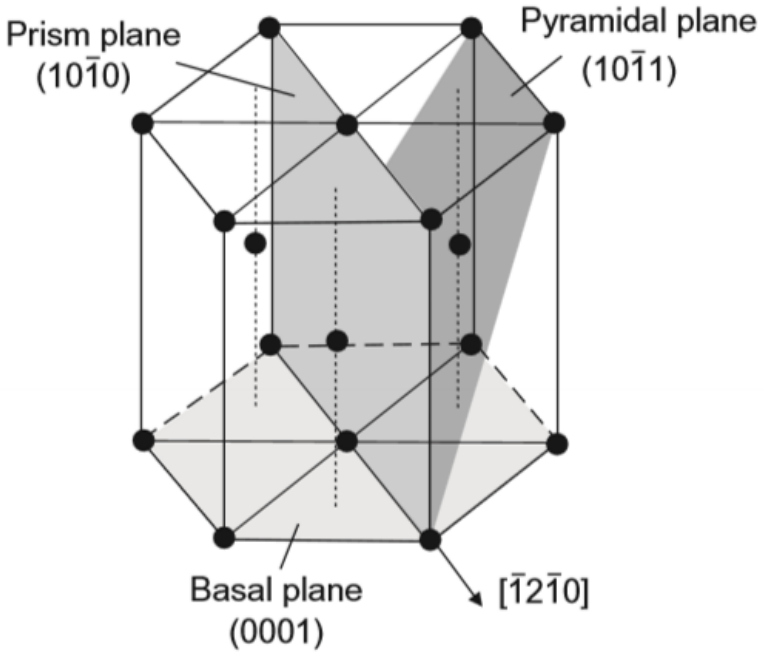
- **Basal (0001)** in Mg, Zn
- **Prism {10$\bar{1}$0}** in Ti, Zr

•Burgers vector often **<11$\bar{2}$0> (picture 1-b)**

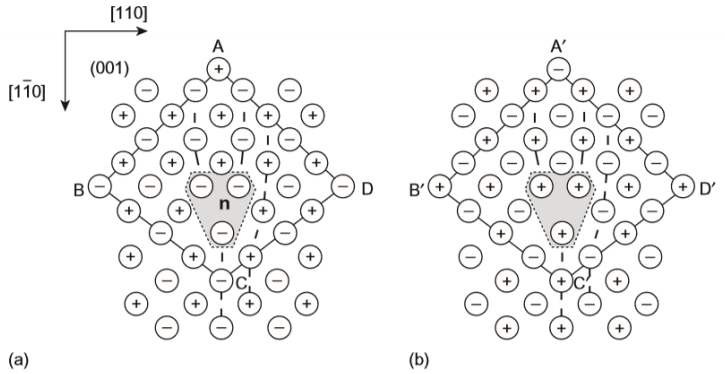
•Dislocations **change into partials**,

•Cross-slip is possible but limited by plane-specific core structures

•**Point defects** affect slip and hardening behavior



Dislocations In IONs

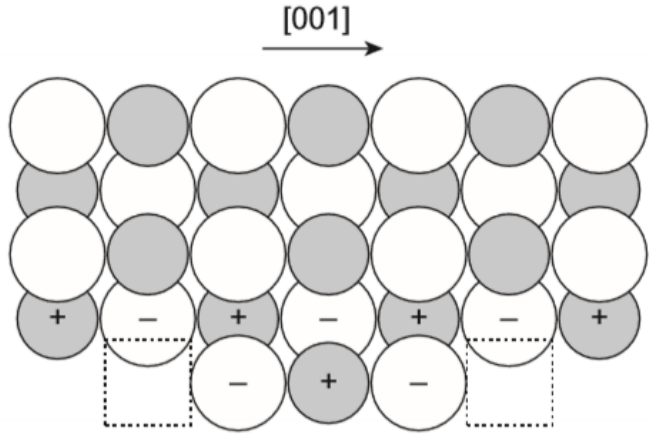


- Materials composed of oppositely charged ions, formed by electron transfer (e.g., NaCl). Dislocations in these crystals can have associated electrical charges, impacting their movement and properties.

•Ionic Crystals (e.g., NaCl, MgO)

•Slip occurs mainly on **{110} planes**; also {100}, {111}, {112} under stress

•Burgers vector: **1/2<110>**



•Dislocations can carry an **effective electrical charge**

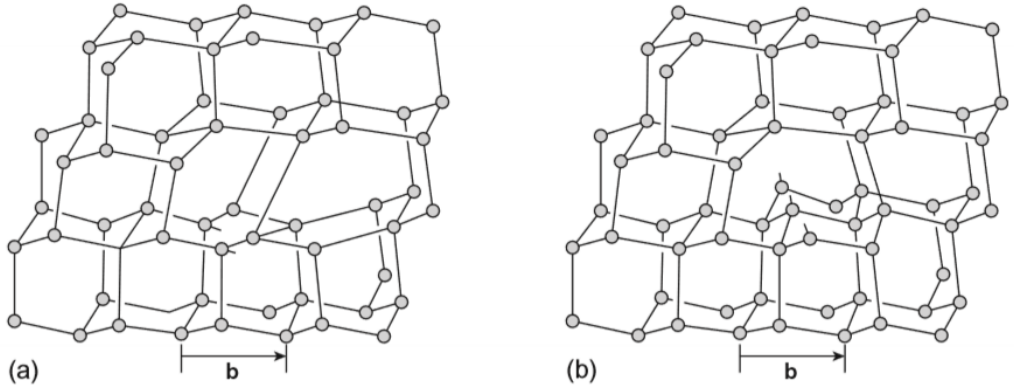
•Charge causes **Coulomb interaction** with vacancies/impurities

•**Dislocation loops** form via vacancy platelets or prismatic punching

•**Hardening** via both **elastic and electrostatic interactions**

Dislocations In Covalants

- Materials with strong, localized, and directional covalent bonds (e.g., silicon, germanium, diamond). These bonding characteristics significantly influence the atomic structure of dislocation cores and their mobility.

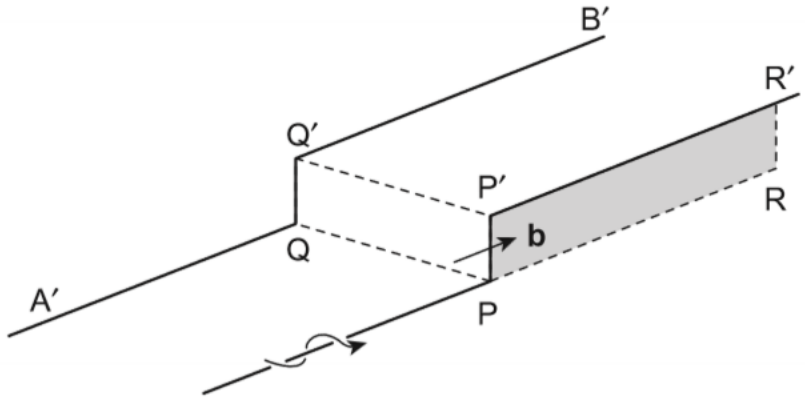


Covalent Crystals (e.g., Si, Ge, Diamond)

- Slip on **{111} planes**, Burgers vector: **1/2<110>**
- Dislocations **dissociate into Shockley partials**
  - Stacking fault energy (~50 mJ/m²) allows TEM visibility
- **High Peierls stress (~10^{-2} G)** due to strong directional bonding
- **Bond reconstruction** (rebonding dangling bonds) lowers core energy
- **Core structures** can be planar or reconstructed (shuffle/glide sets)
- Dislocations affect both **mechanical** and **electrical** behavior
  - **Dangling bonds** trap conduction electrons

Jogs,Superjogs (Specific kind of dislocations)

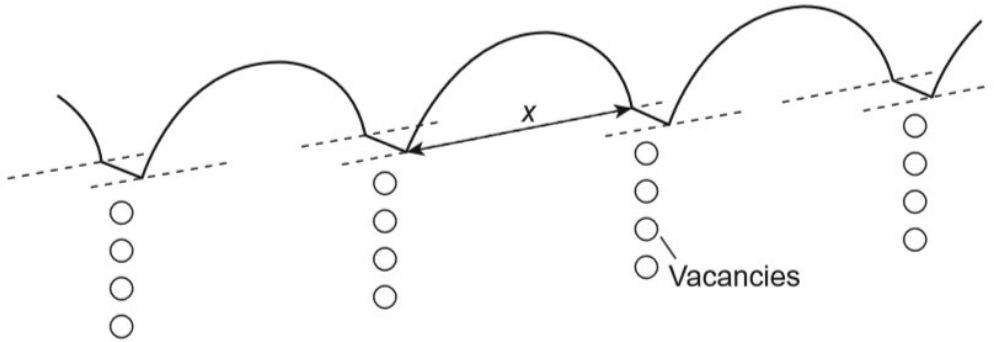
Kind of **dislocations**
which is **like a step**
Depends on the **direction of the Burgers vector**



VACANCIES:

Specifically, **in screw dislocations**,
Jogs can make **vacancies**
or **additional atoms** in crystals (according to Burgers vector)

But in most **metal** cases
The **force** of the atoms controls the jogs and prevents them
from **making vacancies**



Dislocation Origin and Multiplication

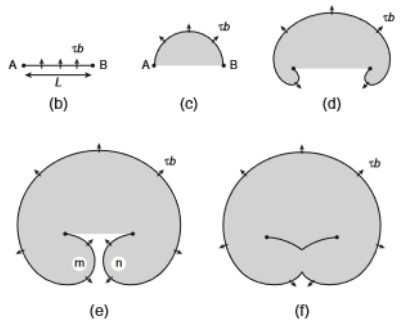
1.Inherited ones(seed crystlas)

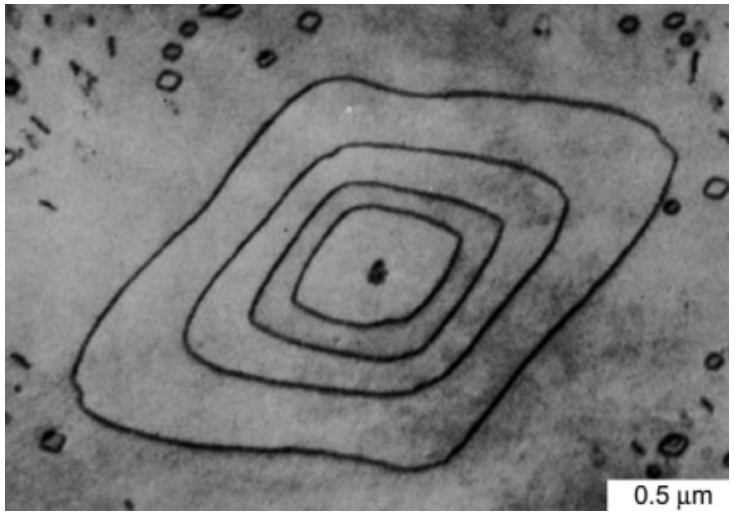
2. Randomly produced(environment effects)

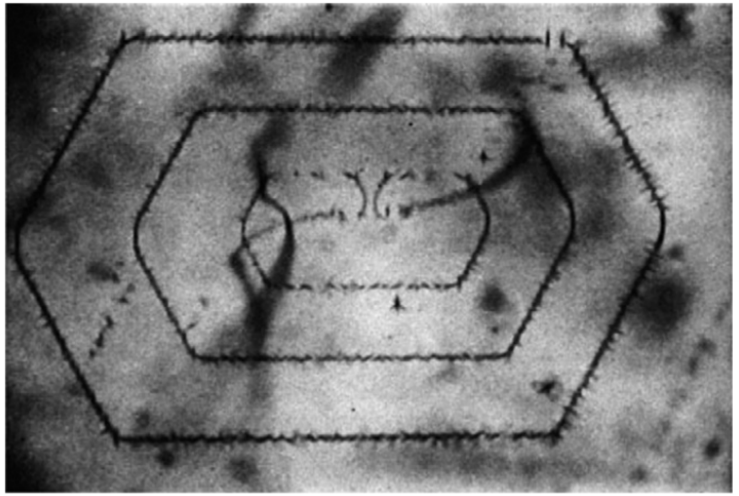
3. Collision of growing crystals(like mountains)

4. Collapse of vacancies(some times→ rappid cooling)

Frank-Read source:





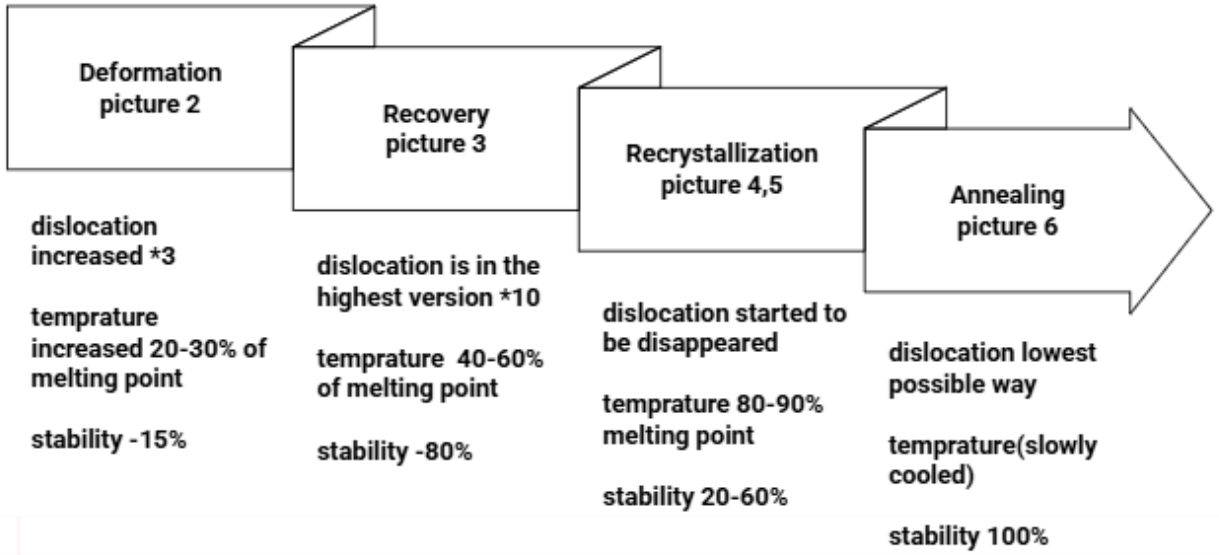


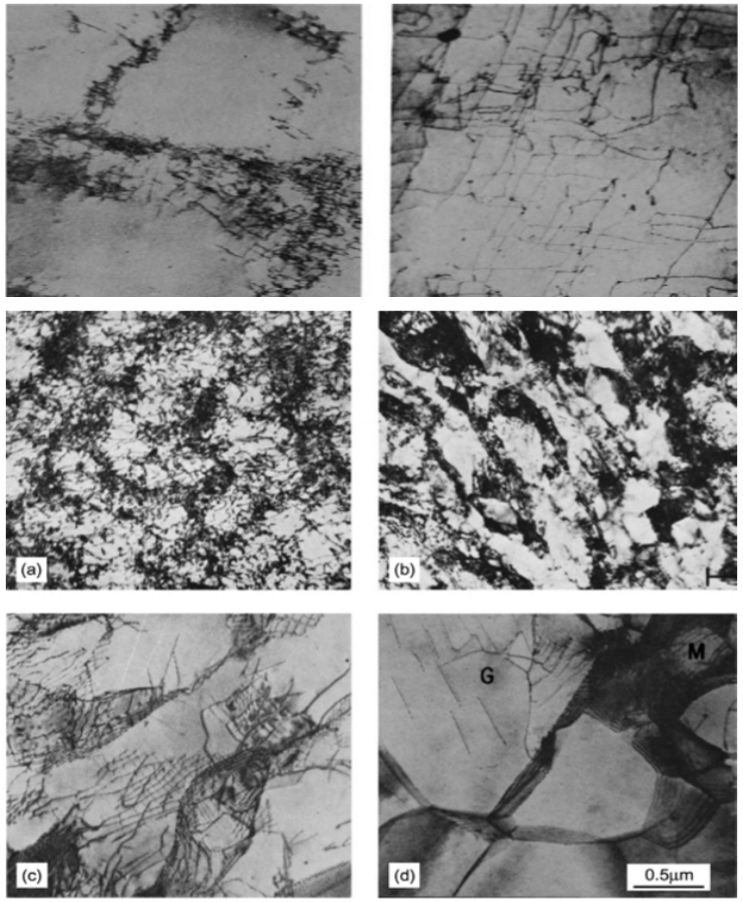
Strengthening and temperatures

Temperature can change:

1. Stability
2. Movement
3. Distribution

Also, by changing the temperature, we have 4 steps:
(All the numbers made by me just for explanation)





- Thanks for your attention!